

SIMATS ENGINEERING

ASSESSMENT TOOL 2: Scenario-Based

COURSE NAME: COMPUTER VISION

COURSE CODE: ITA05

COURSE OUTCOME COVERED

CO2: *Apply image enhancement and segmentation techniques for extracting meaningful regions from images. (BTL 3)*

Assessment Tool Weightage: 20%

1. LOW CONTRAST IMAGE

Scenario: A grayscale image appears dull with very little difference between object and background intensities.

(a) Interpretation and Cause

The image has low contrast because the intensity values of the object and the background are very close to each other. The gray-level range is narrow, so the image looks faded and lacks clear separation between regions.

Causes:

- Poor illumination during image acquisition
- Incorrect camera exposure
- Limited sensor dynamic range
- Compression or transmission effects

(b) Key Issues Affecting Image Visibility

- Poor object-background separation
- Weak edge visibility
- Loss of fine details
- Flat and dull appearance
- Difficulty in human interpretation and computer-based analysis

(c) Appropriate Enhancement Technique

Histogram Equalization

(d) Justification

Histogram equalization redistributes the pixel intensity values across the full gray-level range (0–255). This increases the contrast between dark and bright regions and improves the visibility of hidden details.

(e) Structured Explanation

Step	Explanation
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Problem	Low contrast due to a narrow intensity range
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Issues	Weak edges, poor visibility, loss of detail
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Technique	Histogram Equalization
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Reason	Expands intensity distribution and improves contrast
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Result	Clearer image with better object-background distinction
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Final Conclusion: Histogram equalization is an effective method for enhancing low-contrast grayscale images because it improves brightness variation and reveals important image details.

2. UNEVEN ILLUMINATION

Scenario: An image captured under non-uniform lighting shows bright and dark patches.

(a) Interpretation and Impact

The image suffers from uneven illumination, meaning different parts of the image receive different amounts of light. Some regions appear too bright, while others appear too dark, making important details difficult to observe.

(b) Factors Causing Intensity Variation

- Uneven distribution of the light source
- Shadows from surrounding objects
- Reflections and glare
- Variations in sensor response
- Different distances between the light source and the object

(c) Suitable Enhancement Method

CLAHE (Contrast Limited Adaptive Histogram Equalization)

(d) Justification

CLAHE performs local contrast enhancement by processing small regions independently. It improves dark areas without excessively brightening already bright regions and also prevents excessive noise amplification.

(e) Structured Explanation

Step	Explanation
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Problem	Bright and dark patches caused by non-uniform lighting
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Step	Explanation
Issues	Hidden details, inconsistent brightness, reduced visibility
Technique	CLAHE
Reason	Enhances contrast locally and balances illumination
Result	More uniform brightness and improved detail visibility

Final Conclusion: CLAHE is highly suitable for correcting uneven illumination because it normalizes local brightness while preserving important image details.

3. SALT-AND-PEPPER NOISE

Scenario: A scanned document contains salt-and-pepper noise affecting readability.

(a) Scenario and Characteristics

Salt-and-pepper noise appears as random black and white pixels scattered across the image. It is a type of impulse noise commonly produced by scanner defects, dust particles, faulty sensors, or transmission errors.

(b) Key Issues Affecting Image Quality

- Text becomes difficult to read
- Characters appear broken
- Fine details are corrupted
- False black and white pixels reduce clarity
- OCR accuracy decreases

(c) Appropriate Filtering Technique

Median Filter

(d) Justification

The median filter replaces each pixel with the median value of its neighboring pixels. Since impulse noise contains isolated extreme values, the median operation removes these outliers effectively while preserving edges and text boundaries better than mean or averaging filters.

(e) Structured Explanation

Step	Explanation
Problem	Random black and white pixels in the scanned document
Issues	Broken characters, poor readability, corrupted details
Technique	Median Filter
Reason	Removes impulse noise while preserving edges
Result	Cleaner document with improved readability

4. BLURRED EDGES

Scenario: An image processed for noise removal appears overly smooth, causing edges to blur.

(a) Interpretation and Cause

Excessive smoothing during noise removal suppresses not only noise but also the high-frequency components that represent edges and boundaries. As a result, object outlines become blurred and less distinct.

(b) Key Issues Related to Smoothing and Edge Loss

- Loss of edge sharpness
- Weak boundary definition
- Reduced feature visibility
- Merging of neighboring regions
- Lower accuracy in feature extraction

(c) Appropriate Edge Restoration Technique

Unsharp Masking

(d) Justification

Unsharp masking enhances edges by adding a scaled version of the high-frequency details back to the original image. This restores edge contrast and improves sharpness without introducing excessive noise.

(e) Structured Explanation

Step	Explanation
Problem	Over-smoothing has blurred important edges
Issues	Weak boundaries, reduced sharpness, loss of detail
Technique	Unsharp Masking
Reason	Restores high-frequency edge information
Result	Sharper and clearer object boundaries

5. HIGH-FREQUENCY NOISE

Scenario: A satellite image contains fine-grained noise affecting feature visibility.

(a) Interpretation and Description

High-frequency noise consists of rapid random intensity variations distributed throughout the image. In satellite images, this noise appears as fine-grained texture that hides important geographical and structural features.

(b) Key Issues Affecting Image Clarity

- Reduced visibility of important features
- Appearance of false textures

- Difficulty in edge detection
- Reduced accuracy of object identification
- Overall degradation of image quality

(c) Appropriate Frequency Domain Filtering Method
Gaussian Low-Pass Filter (Frequency Domain)

(d) Justification

A Gaussian low-pass filter gradually attenuates high-frequency components instead of removing them abruptly. This smooth transition effectively suppresses fine-grained noise while minimizing ringing artifacts and preserving the major structures present in the satellite image.

(e) Structured Explanation

Step	Explanation
Problem	Fine-grained high-frequency noise in the satellite image
Issues	False textures, poor edge detection, reduced clarity
Technique	Gaussian Low-Pass Filter
Reason	Smoothly suppresses high-frequency noise while preserving important structures
Result	Cleaner image with improved feature visibility

Code of Conduct:

I _____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

RUBRICS FOR CALCULATION:

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)

Understa nding of Scenario	20%	Demonstrates clear and complete understandin g of the scenario and context	Shows good understandin g with minor gaps	Partial understandi ng; misses key aspects	Misinterprets or fails to understand the scenario
Identificat ion of Key Issues	20%	Accurately identifies all relevant issues and factors involved	Identifies most key issues with minor omissions	Identifies limited issues; some irrelevant points	Unable to identify key issues
Applicatio n of Concepts	25%	Applies appropriate concepts effectively to address the scenario	Applies concepts with minor errors	Limited or partially correct application	Unable to apply relevant concepts
Decision Making	20%	Makes well- reasoned, logical decisions with strong rationale	Makes reasonable decisions with some justification	Makes weak decisions with limited justification	No clear decisions made or rationale provided
Clarity of Respons e	15%	Response is clear, structured, and logically presented	Generally clear with minor issues	Somewhat unclear or poorly structured	Disorganize d and difficult to understand